ISO 26262: functional safety for road vehicles

ISO 26262 is the international standard for the functional safety of electrical and electronic systems in series-production road vehicles. It adapts the general standard IEC 61508 to the automotive life cycle, from concept and development to production, operation and decommissioning.

Hazard analysis and risk assessment (HARA) is the starting point. Each hazardous event is rated for severity (S0 to S3), exposure (E0 to E4) and controllability (C0 to C3). The combination gives the Automotive Safety Integrity Level, or ASIL.

Automotive Safety Integrity Levels

There are four ASILs, from ASIL A (lowest) to ASIL D (highest). Hazards that need no safety measures beyond normal quality management are classified as QM. An airbag or a braking system is typically ASIL D, while rear lights are often ASIL A or B.

A higher ASIL demands stricter methods: more rigorous requirements tracing, more testing, higher coverage targets and more independence in reviews. ASIL decomposition allows a high-ASIL requirement to be split across redundant, independent elements with lower ASILs.

The V-model

The V-model structures development: requirements and design on the left side, implementation at the bottom, and verification and validation on the right side, where each test level checks the matching design level. Unit tests check the software units, integration tests check how components work together, and vehicle-level validation checks that the safety goals are met.